

[DC] Towards Universal VR Sickness Mitigation Strategies

Isayas Berhe Adhanom *

Department of Computer Science and Engineering
University of Nevada, Reno

ABSTRACT

Although various issues related to VR hardware and virtual content have been identified as likely causes of VR sickness, human factors, such as age, gender and personal susceptibility to VR sickness, determine whether a user will experience VR sickness or how severe their symptoms will be. However, current VR sickness mitigation or measurement techniques do not take these human factors into consideration. This doctoral thesis aims to identify and quantify the human factors that cause difference in susceptibility to VR sickness among different demographic user groups.

Index Terms: Human-centered computing Virtual Reality

1 INTRODUCTION AND MOTIVATION

Though consumer VR headsets have significantly advanced in recent years, VR sickness is still considered a major barrier to the mass adoption of VR. While up to 67% of adults may experience mild to severe symptoms [4], there is substantial evidence that some users are more susceptible to VR sickness than others, for example, there is ample evidence that women are more likely to experience VR sickness than men [7, 13], and VR sickness has been considered a likely contributing factor to the low adoption rate of VR in women [2].

There are currently various theories that aim to explain VR sickness, including the sensory conflict theory [14], the postural instability theory [15] and the eye movement theory [6]. However, the sensory conflict theory, which attributes VR sickness to a conflict between the visual, vestibular, and proprioceptive senses, seems to be the most widely accepted theory [8].

Causes of VR sickness could be divided into three domains: hardware, content or human factors [5]. Hardware factors, include issues related to the display type, the field of view (FOV) of the display device, the refresh rate, and display latency, whereas content factors include issues related to the virtual content such as optical flow experienced by users, the type of virtual locomotion used and graphic fidelity of the virtual scene. Although the type of hardware used and the characteristics of the virtual content are major causes of VR sickness, human factors including gender, age and individual susceptibility to VR sickness determine whether a specific user will suffer from VR sickness, and if so what the severity of the VR sickness will be.

However, the most popular approaches to VR sickness reduction focus on the hardware and content factors. Popular VR sickness mitigation efforts such as reducing the FOV, providing a reference frame, minimizing latency and others rarely consider human factors while implementing VR sickness mitigation mechanisms.

These sickness reduction systems in which one VR sickness reduction strategy is used for all user groups without considering their special user or demographic profile might be easier and cheaper to implement, but they could have negative or unintended consequences on some users. For example, FOV restriction is a widely

used VR sickness reduction strategy, but reducing the FOV could have a negative effect on women since women have a wider field of view.

In order to ensure VR sickness reduction strategies are effective for all user and to minimize unintended side effects from these strategies to specific demographics, it is necessary to develop and evaluate VR sickness mitigation or measurement strategies that take into consideration the human factors involved in inducing or exacerbating VR sickness. To this end, my research focuses on identifying the factors that cause differences in susceptibility to VR sickness, and developing or improving VR sickness mitigation strategies to make them universally effective for all demographic groups.

2 CURRENT PROGRESS

We have already conducted two studies to evaluate how FOV restriction - the most widely used VR sickness mitigation strategy - could affect women differently compared to men. Then based on our findings from these two studies we developed an adaptive foveated FOV restrictor that solves some of the issues with current FOV restriction strategies.

In our first study [2], we investigated whether dynamic FOV restriction affects path integration ability differently in women versus men. Path integration is one component of spatial navigation and refers to the process of integrating self-motion (using inertial/visual/auditory cues) over time to obtain an estimate of one's current position relative to a starting point [11]. Our results showed, contrary to our expectations, that FOV restriction did not impede path integration ability in men or women. However, a limitation in our first study was that path integration is considered to be only one component of spatial navigation which also involves skills like spatial learning, spatial memory, and constrained route planning [16]. For spatial navigation, women reportedly rely more on landmarks to cognitively map spaces [17]. Applying an FOV restrictor can block peripheral landmarks which might impede spatial learning to a larger extent than path integration ability.

In our second study (in press at Transaction of Applied Perception) we specifically explored spatial learning and how it is affected by FOV restriction given that no studies have investigated this yet. In this study, we used the Morris Water Maze (MWM) to evaluate spatial learning and memory of our participants. First introduced by Richard Morris in 1981 [12], the Morris Water Maze (MWM) task has been one of the gold standards in behavioral neuroscience to evaluate spatial learning and memory in rodents [3]. Studies on human spatial learning and memory have used a virtual version of the MWM [3]. Our study found that a gender difference in spatial learning ability exists between men and women.

From our first two studies, we were able to identify that existing FOV restrictor implementations use a head-fixed restrictor where the effect is applied to the center of the head-fixed FOV which is only updated by the user's head gaze. With this approach, peripheral motion stimulation is only optimally blocked when the head and eye gaze align, i.e, when the user is looking at the center of the HMD. However, this may not always be the case; for example, the user's eye gaze may shift to eccentric targets in the visible field that are blocked by the restrictor. In such a case, the effectiveness of the

*e-mail: iadhanom@nevada.unr.edu

FOV restrictor is impeded because the user's peripheral vision is not fully blocked by the restrictor exposing it to optical flow, which could increase VR sickness, and the restrictor could fall on the users foveal region, which could break immersion. These issues have the potential to affect women more than men, as women have a wider FOV [9, 10].

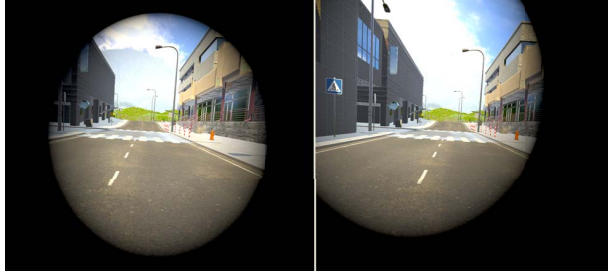


Figure 1: An adaptive foveated FOV restrictor (right) follows the users eye gaze and allows them to see more of the scene while still blocking optical flow in the periphery. The user is looking at the top left of the scene in both images. Notice that the road sign is visible only when using the adaptive FOV restrictor.

To solve these issues, in our third study [1] we developed a new type of FOV restrictor, one where the restrictor is gaze responsive. This foveated FOV restrictor assures that peripheral vision always remains blocked from any optical flow, which could lower VR sickness, and makes it impossible for a user to look at the restrictor itself, allowing users to visually explore a larger portion of the virtual environment during locomotion. Our evaluation of the new system showed that there was a significant difference in eye gaze behavior, as measured by eye gaze dispersion, with the foveated FOV restrictor allowing participants to have a wider visual scan area compared to the head-fixed FOV restrictor (the widely used approach), which confined their eye gaze to the center of the FOV. Although it needs further research, this has the potential to increase spatial navigation performance of users, especially women.

3 FUTURE RESEARCH GOALS

Current content based VR sickness mitigation or measurement techniques rarely take into consideration the user's specific personal or demographic profile (i.e. gender, age etc). Previous studies show that there are differences in eye movements, vestibular function, and sensory integration between various demographic groups, and could be the main causes of differences in susceptibility to VR sickness. The development of new VR sickness mitigation or measurement strategies should consider these differences to be universally effective across all demographic groups. Considering these limitations, we believe there is still plenty of room left for further research into human factors and their relations to VR sickness.

4 RESEARCH QUESTIONS

The three research questions that we are specifically interested are:

- **RQ1:** Does VR cause differences in eye movements between different demographic groups, such as men vs women or children vs adults, and are they related to VR sickness?
- **RQ2:** Do differences exist in sensitivity to visual self motion (vection) during locomotion between different demographic groups?
- **RQ3:** Do differences in sensory integration during virtual locomotion relate to VR sickness susceptibility?

5 CONCLUSION

My PhD research is dedicated to developing universal VR sickness mitigation strategies that take into consideration the human factors

involved in inducing or exacerbating VR sickness. In the past two years I have conducted research that aims to understand how FOV restriction - a popular VR sickness strategy - affects women's task and spatial navigation performance in VR. I plan to continue my research to identify other human factors that cause differences in VR sickness susceptibility between different demographic groups. To this end I have identified three research questions that I want to further investigate. As future work, I contemplate the following open questions: are these research questions adequate to address the differences in susceptibility to VR sickness between different user groups, are there other human factors that we need to look at?

REFERENCES

- [1] I. B. Adhanom, N. Navarro Griffin, P. MacNeilage, and E. Folmer. The Effect of a Foveated Field-of-view Restrictor on VR Sickness. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, number February, pp. 645–652. IEEE, 3 2020. doi: 10.1109/VR46266.2020.1581314696458
- [2] M. Al Zayer, I. B. Adhanom, P. MacNeilage, and E. Folmer. The effect of field-of-view restriction on sex bias in vr sickness and spatial navigation performance. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, pp. 1–12, 2019.
- [3] R. S. Astur, M. L. Ortiz, and R. J. Sutherland. A characterization of performance by men and women in a virtual morris water task: A large and reliable sex difference. *Behavioural brain research*, 93(1-2):185–190, 1998.
- [4] C.-H. Chang, W.-W. Pan, L.-Y. Tseng, and T. A. Stoffregen. Postural activity and motion sickness during video game play in children and adults. *Experimental brain research*, 217(2):299–309, 2012.
- [5] E. Chang, H. T. Kim, and B. Yoo. Virtual reality sickness: a review of causes and measurements. *International Journal of Human-Computer Interaction*, 36(17):1658–1682, 2020.
- [6] S. M. Ebenholtz. Motion sickness and oculomotor systems in virtual environments. *Presence: Teleoperators & Virtual Environments*, 1(3):302–305, 1992.
- [7] D. A. Graeber and K. M. Stanney. Gender differences in visually induced motion sickness. In *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, vol. 46, pp. 2109–2113. SAGE Publications Sage CA: Los Angeles, CA, 2002.
- [8] B. Keshavarz, H. Hecht, and B. Lawson. Visually induced motion sickness: characteristics, causes, and countermeasures. *Handbook of Virtual Environments: Design, Implementation, and Applications*, pp. 648–697, 2014.
- [9] E. M. Kolasinski. Simulator sickness in virtual environments. Technical report, DTIC Document, 1995.
- [10] J. J. LaViola Jr. A discussion of cybersickness in virtual environments. *ACM SIGCHI Bulletin*, 32(1):47–56, 2000.
- [11] J. M. Loomis, R. L. Klatzky, and R. G. Golledge. Navigating without vision: basic and applied research. *Optometry and Vision Science*, 78(5):282–289, May 2001.
- [12] R. G. Morris. Spatial localization does not require the presence of local cues. *Learning and motivation*, 12(2):239–260, 1981.
- [13] J. Munafo, M. Diedrick, and T. A. Stoffregen. The virtual reality head-mounted display occlus rift induces motion sickness and is sexist in its effects. *Exp. Brain Research*, pp. 1–13, 2016.
- [14] J. T. Reason and J. J. Brand. *Motion sickness*. Academic press, 1975.
- [15] G. E. Riccio and T. A. Stoffregen. An ecological theory of motion sickness and postural instability. *Ecological psychology*, 3(3):195–240, 1991.
- [16] J. J. Rieser, D. Guth, and E. Hill. Mental processes mediating independent travel: Implications for orientation and mobility. *Journal of Visual Impairment and Blindness*, 1982.
- [17] D. G. Woolley, B. Vermaercke, H. O. de Beeck, J. Wagemans, I. Gantois, R. D'Hooge, S. P. Swinnen, and N. Wenderoth. Sex differences in human virtual water maze performance: Novel measures reveal the relative contribution of directional responding and spatial knowledge. *Behavioural brain research*, 208(2):408–414, 2010.